Answer Key

1. $t_1 = 2\sqrt{\frac{2h_1}{g}}$

$t_2 = 2\sqrt{\frac{2h_2}{g}}$

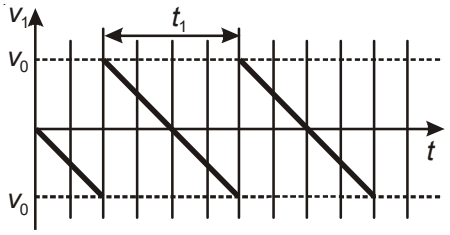


Fig. 1

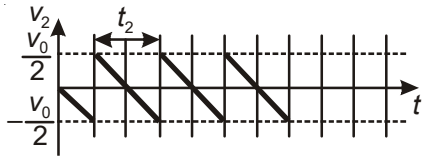


Fig. 2

There are two possibilities for the coincidence of the velocities of the balls in magnitude and direction. The velocities of the balls may coincide for the first time $\tau = nt_1$ s after the beginning of motion (where n = 0, 1, 2, ...) during the time interval $\frac{t_1}{4}$ then they coincide $\frac{3t_1}{4}$ s after the beginning of motion during the time interval $\frac{t_1}{2}$. Subsequently, the velocities will coincide with a period t_1 during the time interval $\frac{t_1}{2}$. The other possibility consists in that the second ball starts moving $\tau = \frac{t_1}{2} + nt_1$ s (where n = 0, 1, 2, ...) after the first ball. After $\frac{t_1}{4}$ s, the velocities of the balls coincide for the first time and remain identical during the time interval $\frac{t_1}{2}$. Subsequently, the situation is repeated with a period t_1.

2. $\frac{nr\cos\alpha}{v} = k\sqrt{\frac{2H}{g}}$.

3. $\omega = \frac{v}{R+r}\sin\alpha \cdot \tan\frac{\alpha}{2} = \frac{2v\sin^2(\alpha/2)}{(R+r)}$.

4. $t_2 = \left(\sqrt{5}-1\right)t_1$.

5. $v_2 = \dfrac{v_1}{\cos\alpha} - v_1 = v_1\dfrac{2\sin^2(\alpha/2)}{\cos\alpha}$ and is directed upwards.

6. $v_0 = \dfrac{vR}{R\sin\alpha - r}$.

7. The velocities of the points of the ingot lying on a segment *AB* at a given instant uniformly vary from v_1 at point *A* to v_2 at point *B*. Consequently, the velocity of point *O* (Fig. 7) at a given instant is zero.

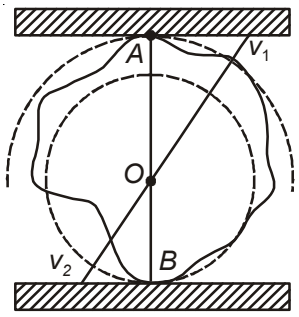


Fig. 7

Hence point *O* is an instantaneous centre of rotation. (Since the ingot is three-dimensional, point *O* lies on the instantaneous rotational axis which is perpendicular to the plane of the figure.) Clearly, at a given instant, the velocity v_2 corresponds to the points of the ingot lying on the circle of radius *OA*, while the velocity v_2 to points lying on the circle of radius *OB*. (In a three-dimensional ingot, the points having such velocities lie on cylindrical surfaces with radii *OA* and *OB* respectively.)

8. $k \approx 2k_0\cos^2\alpha$.

9. Consequently, the left load will be the first to touch the floor.

10. $m_2 < m_1\cos 2\alpha$.

11. $N = \dfrac{mg}{\sqrt{2}} - \dfrac{mv^2}{2r}$.

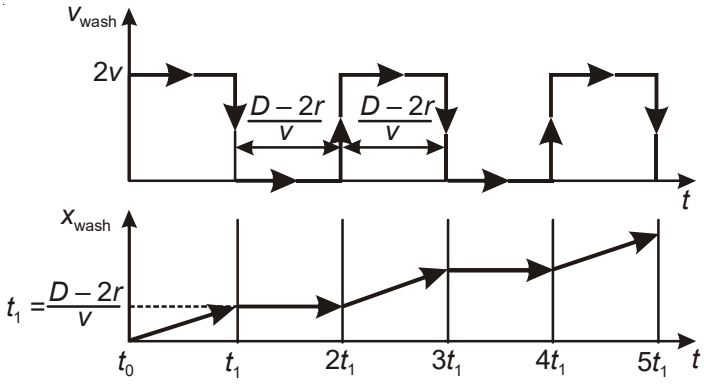
12. $\mu \geq \dfrac{1}{3}\left|\dfrac{1}{\tan\alpha} - \dfrac{2}{\tan\beta}\right|$.

13. The smaller disc will rotate at the steady-state angular velocity Ω..

$M = \mu mgd$

must be applied to the axis of the larger disc.

14.



15. $a = \dfrac{1}{1+4(M/m)\tan^2\alpha} g.$

16. As a result, we find that $u = v/\sqrt{2}$ and $v' = 0$. Consequently, the velocities of the extreme blocks after the impact form the angles of 45° with the velocity v, the washer stops, and the middle block remains at rest.

17. $v_0^2 \geq \dfrac{g'r}{\sin^2\phi} = \dfrac{gr\cos\alpha}{\sin^2\phi}.$

18. $T = \dfrac{mv^2 l}{2r^2}\left(1+\dfrac{gl^2}{4rv^2}\right) \approx \dfrac{mv^2 l}{2r^2}.$

19. $v \leq \sqrt{gr(3\cos\alpha - 2)}$

20. $l = \sqrt{2vrt}.$

21. The left block reaches the pulley before the right block strikes the table

22. At any instant the right load is lower than the left one.

23. $\omega = g^{1/2}\left(\dfrac{m_1^2 - m_2^2}{m_1^2 l_1^2 - m_2^2 l_2^2}\right)^{1/4} \simeq 14 \text{ rad/s}.$

24. $\mu_{min} = \dfrac{\sqrt{2}}{4}.$

25. $\dfrac{T_1}{T} = 1 - \dfrac{1}{\pi}\arccos\dfrac{\beta}{\alpha}.$

26. $T = 2\pi\sqrt{\dfrac{\sqrt{3}\ell}{10g}}$

27. $T = 2\pi\sqrt{\dfrac{\ell'}{g'}} = 2\pi\sqrt{\dfrac{2\ell}{g}}.$

28. $T = 2\pi\sqrt{\frac{\ell_1\ell_2}{ag}}$.

29. $h = r$

30. $a = \frac{g(h_2 - h_1)}{h_2 + h_1}$.

31. $\alpha \geq 2$

2. HEAT AND MOLECULAR PHYSICS

1. $A = nR(T_2 - T_1)\left(\frac{T_2}{T_1} + \frac{T_3}{T_2} - 2\right)$.

2. The amount of heat received by the gas in the process $1 \rightarrow 3$ is larger.

3. $Q' = Q - 3nRT - \frac{5}{2}kl^2$.

4. $C = \frac{\Delta Q}{\Delta T} = 2R$.

5. $p_1 = \frac{p_0 + mg/S}{1 - \Delta h/h}$.

6. $u = a_m t_0 = \frac{p_0 abc}{2mv}$.

3. ELECTRICITY AND MAGNETISM

1. $E = E_0 \sin\frac{\alpha}{2}$.

2. $\phi_2 = \phi_1 \frac{r_1}{r_2}$.

3. $I(t) = \begin{cases} -v\left(\frac{q_1}{a} + \frac{q_2}{b}\right), & t < \frac{a}{v}, \\ -v\frac{q_2}{b}, & \frac{a}{v} \leq t < \frac{b}{v}, \\ 0, & t \geq \frac{b}{v}. \end{cases}$

4. $U = \frac{\sqrt{4\alpha U_0 + 1} - 1}{2\alpha} = 12\ \text{V}$.

5. $d = \dfrac{2\ell q^2}{q^2 + 8\pi\varepsilon_0 mv^2\ell}$.

6. $I = \dfrac{15}{7}\dfrac{U}{R}$.

7. $R_{AB} = \dfrac{15}{11}R$.

8. $R_x = \dfrac{\varepsilon_1}{I}$.

9. $q_{st} = \dfrac{C(\varepsilon_2 R_1 + \varepsilon_1 R_2)}{R_1 + R_2}$.

10. $x_2 = 0.08d$.

11. $R_2 = rR_1/(r + R_1)$

12. $t = \dfrac{2m}{eB}\arctan\left(\dfrac{eBR}{mv}\right)$.

13. The coordinate of the loop in a certain time t after the beginning of motion will be

$$z = \frac{mgL}{a^4\alpha^2}\left[-1 + \cos\left(\frac{a^2\alpha}{\sqrt{Lm}}t\right)\right].$$

14. $q_0 = \dfrac{CU}{2}\sqrt{2 - \cos^2\omega_0\tau}$.

4. OPTICS

1. Rays which are singly reflected from the mirror surface of the cone propagate as if they were emitted by an aggregate of virtual point sources arranged on a circle. Each such source is symmetrical to the source S about the corresponding generator of the cone. The image of these sources on a screen is a ring. It is essential that the beam of rays incident on the lens from a virtual source is plane: it does not pass through the entire surface of the lens but intersects it along its diameter. Therefore, the extent to which such a beam is absorbed by a diaphragm depends on the shape and orientation of the latter.

   A symmetrical annular diaphragm (Fig. 1) absorbs the beams from all virtual sources to the same extent. In this case, the illuminance of the ring on the screen will decrease uniformly. The diaphragm shown in Fig. 2 will completely transmit the beams whose planes form angles $\alpha < \alpha_0$ with the vertical. Consequently, the illuminance of the upper and lower parts of the ring on the screen will remain unchanged. Other beams will be cut by the diaphragm the more, the closer the plane of a beam to the horizontal plane. For this reason, the illuminance of the lateral regions of the ring will decrease as the angle α varies between α_0, and $\dfrac{\pi}{2}$.

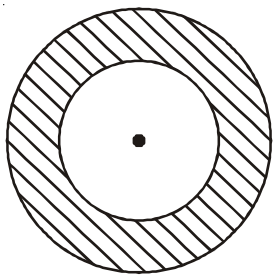


Fig. 1

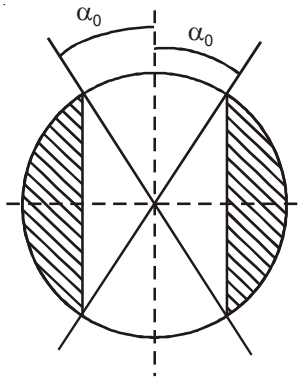


Fig. 2

2. $L = L_0 \dfrac{2(n_{g\ell} - 1)n_w}{2n_{g\ell} - n_w - 1} = 30$ cm.

3. $t = \dfrac{2h}{v}\left(\sqrt{1 - n^2 \sin^2 \dfrac{\alpha}{2}} + \dfrac{\sqrt{l^2 - h^2}}{h} n \sin\dfrac{\alpha}{2}\right)$.

4. It follows from symmetry considerations that the image of the point source *S* will also be at a distance *b* from the sphere, but on the opposite side

5. The easiest way is to determine the displacement of the visible contour of the face behind the glasses relative to other parts of the face: if it is displaced inwards, the lenses are diverging, and your companion is short-sighted, if it is displaced outwards, the lenses are converging, and the person is longsighted.

6. $\Delta x = x_2 - x_1 \simeq 0.6$ cm

Anskwer Key

1. Cylinder will blow up at 1200 K before its M.pt.
2. $P_{(final)}$ = 0.5714 atm

   Mole of H_2 in flask I = 0.4

   Mole of H_2 is flask II = 0.3
3. P_{mix} = 1.158 atm

   Mole % of N_2 = 31.8%

   Mole % of O_2 = 68.2%
4. Number of Hg-atom = 6324 atoms per litre
5. 36.51cm

   85.25 cm of Hg
6. (a) 98.03% (N_2), 1.97% of H_2O

   (b) 10.16 L
7. Moles (CO_2) = 3.1×10^{-8}

   Moles (H_2O) = 1.7×10^{-8}

   Moles (N_2) = 2.1×10^{-8}
8. Mole % of N_2 = 35.33

   Mole % of NO = 38.67

   Mole % of NO_2 = 26.00
9. 0.289 torr/sec
10. $\frac{r_{UF_6}^{235}}{r_{UF_6}^{238}} = 1.0043$ & 638 steps
11. 725 mg
12. $P_n = P\left[\frac{V}{V+v}\right]^n$
13. 227 K
14. (c) 300 K
15. (a, b, c)
16. (a, b, c, d)
17. NH_3
18. Mass of CH_4 = 0.8 g

    Volume of O_2 = 2.24 L
19. 0.2 cm^2
20. 122.31
21. $X_{(N_2O_5)} = 0.4$

22 470 mL

23. (6)

24. (2)

25. (4)

26. (c)

27. 48×10^{-42} m^3

28. 21.87×10^4 J

29. n = 6, Z = 3

30. $\Delta p = 5.27 \times 10^{-22}$ Ns

31. Z = 24
    Element -Chromium

32. 13%

33. (c)

34. 16.423 MeV

35. 3.12 MWg^{-1}

36. 95.8%

37. 60.08 kJ

38. P_{Total} = 2.887 atm
    P(He) = 1.443 atm

39. Avogadro's No = 6.775×10^{23} particle/mol

40. r_A = 0.0626 μCi
    r_B = 0.0313 μCi
    Molar ratio after 2 days = 1.5

41. E = 5.816×10^{-13} J
    $\nu = 8.77 \times 10^{20}$ Hz
    $\lambda = 3.4 \times 10^{-13}$ m

42. 7.56×10^{-3} M

43. 2.85×10^{20} dph

44. (2)

45. (d)

46. $8Cl_2 \rightarrow 2Cl^{7+} + 14Cl^-$

47. (b)

48. (a, b, c)

49. % error in ammeter = 2.05%

50. Blue color

51. w_{cu} = 4.763 g
    i = 0.804 ampere

52. $[Cu^{2+}]$ = 0.094 M

$[H^+]$ = 0.012 M

$[SO_4^{2-}]$ = 0.1 m

If Cu electrodes are used, no change in molarity of electrolyte i.e = 0.1 M

53. E° = 1.1830 V
54. 265.04 ampere hr
55. pH = 1.1992
56. E° = –0.1658V
57. 99.967%
58. E = 0.458 volt
59. $K_f = 8.23 \times 10^{63}$
60. Cell Reaction : $Hg_2Br_2(s) + 2Ag(s) \rightarrow 2Hg(l) + 2AgBr(s)$

    ΔG = –13.124 KJ ΔH = 9.882 KJ ΔS = 77.2 J
61. (d)
62. (d)
63. (a)
64. (a) 5
65. Strength of $FeSO_4$ = 13.64 g/litre

    Strength of $Fe_2(SO_4)_3$ = 9.45 g/litre
66. $M_{NO_3^-} = \frac{1.37}{8} = 0.1716$
67. Ratio of Fe in $FeSO_4$ and $FeC_2O_4 = \frac{7}{3}$
68. 39.6 g/litre
69. V = 36.36 mL

    If Cu is used, then V = 24.24 mL
70. Vol. % of $O_3 = 1.847 \times 10^{-3}$ %
71. $A_3 (CO_3)_2(OH)_2$
72. % mass of Cu_2O = 94.67
73. 1 : 1
74. (a)
75. (d)
76. (a, b, c)
77. (a, d)
78. $P_{(C_2H_4Br_2)} = 104.9$ mm

    $P_{(C_3H_6Br_2)} = 48.63$ mm

    $P_M = 153.53$ mm
79. P_S = 759.2 mm

    ΔP = 0.21 mm
80. A = 25.59 g/mol

B = 42.64 g/mol

81. $\frac{(\Delta T_f)_A}{(\Delta T_f)_B} = \frac{3}{1}$

$\frac{(\Delta T_f)_{S_1}}{(\Delta T_f)_{S_2}} = 0.81$

82. No. of ions = 3

83. Freezing point = 270.23 K

84. $\Delta H = [313.7 \times 2.303 + 1.40655T] R$
at 80 K, ΔH = 1670 cal

85. (4)

86. (1)

87. (a)

88. (d)

89. (a, c, d)

90. V = 2.386×10^{-5} litre

91. E_a = 2798 J = 2.8 kJ

92. (c)

93. (a)

94. $K_1 = K_2 = 0.046$

95. $E_{AV} = \frac{E}{3}[2n + 1], K_{AV} = \frac{n}{2} \times [n+1]K$

96. Order of Reaction = 2 + 1 = 3

97. $K = 3.27 \times 10^{-3}\ min^{-1}$

98. $\frac{r_0}{r_m} = 0.261$ (P_0 = 0.4 ; P_m = 0.98)

99. K = 0.022 hr^{-1}

100. K = $2.07 \times 10^{-2}\ min^{-1}$

101. (d)

102. (a, b, c, d)

103. (a) $\rightarrow$ (ii); (b) $\rightarrow$ (iii); (c) $\rightarrow$ (i); (d) $\rightarrow$ (iv)

104. Kc = 1.8

105. (a) Moles of O_2 = 0.1
(b) Moles of O_2 = 0.4

106. $P_{H_2S} = 0.387$ atm

107. $K_P = 5.1 \times 10^{11}$

108. Above 1078.5 K, value of K will be greater than 1

109. Overall stability order : B > C > A.

$$CH_3 \cdot CH_2 \cdot CH_2 \cdot C \equiv CH \xrightarrow{KOHalc} [CH_3\,CH_2\,CH = C = CH_2] \longrightarrow [CH_3CH_2C \equiv C - CH_3]$$

110. K_P = 6.14 atm

111. T = 1167.26 K or 894.26°C

112. $K_{C_1} = 0.11$ and $K_{C_2} = 0.142$

113. P = 0.126 atm

114. (1)

115. (d)

116. Case I : $[OH^-] = [NH_2C_2H_4NH_3]^+ = 3.57 \times 10^{-3}$ M

Case II: $[NH_3C_2H_4NH_3]^{2+} = 2.7 \times 10^{-8}$ M $= K_{b_2}$

117. $K_a = 1.2 \times 10^{-5}$

118. $[I^-] = 8.5 \times 10^{-16}$ M

% of Ag precipitated = 99.83%

119. $K_{SP}(BaCO_3) = 4.9 \times 10^{-10}$

120. CdS will be precipitated first.

121. (a) $2NO_2(aq.) + 2H_2O(l) \longrightarrow HNO_2(aq.) + H_3O^+(aq.) + NO_3^-(aq.)$

(b) $[NO_2^-] = 4.5 \times 10^{-4}$

pH = 1.17

(c) π = 4.95 atm

(d) w_{CaO} = 0.924 g

122. pH = 7.81

123. (d)

124. (b)

125. R = 66.67 ohm

126. V = 2372.5 × 10^2 litre

127. $K_{sp} = 7.699 \times 10^{-17}$

128. $V_{(l)} = 2.01 \times 10^5$ litre

129. (b)

130. (a) $\Delta U = 0$

(b) temperature remains constant

131. ΔU = 37.5 kJ mol

ΔH = 40.6 kJ mol

132. Heat produced = 5.375 kJ

Final temperature = 1177.27 K

Final pressure 14.39 atm

133. Athlete should go for 303 lifting.

134. $\Delta S_T = 7522.50$ J

135. $(K_P)_{300} = 8.94 \times 10^4$

$(K_P)_{1200} = 0.77$

At 300 K, reaction will proceed in forward direction and at 1200 K in backward direction

136. (8)

137. (2)

138. (a)

139. (a)

140. (a)

141. (a)

142. (b, d)

143. (a, b, c, d)

144. (a, c)

145. $W_{Hg} = 93.44$ g

146. Case I : 34 days

Case II : 24 days

147. Amount of glucose burnt = 29.25 g

Heat produced = 458.66 kJ

148. Molar heat capacity of liquid = 39.04 J/mol

149. Eq. of NaOH used for neutralisation of HA = 0.685

150.13.4 Kcal

151. (a)

152. (d)

153.62% empty space

154. 3.4772×10^{24} atoms

155. (a) 12

(b) 0.707 a

156. (a)

157. (c)

158. (b)

❑ ❑ ❑

ANSWERS

1.	(2)	2.	(1)	3.	(2)	4.	(2)	5.	(2)
6.	(1, 2)	7.	(2, 3)	8.	(1, 2)	9.	(4)	10.	(2)
11.	(4)	12.	(3)	13.	(2)	14.	(2)		
15.	A(p); B(s); C(q); D(r)								
16.	A(q); B(r); C(s); D(p)								
17.	0	18.	2	19.	7	20.	3	21.	(1)
22.	(4)	23.	(1)	24.	(4)	25.	(1)	26.	(3)
27.	(3)	28.	(1)	29	(3)	30.	(3)	31.	(1, 2, 4)
32.	(1, 3, 4)	33.	(1, 2)	34.	(3)	35.	(3)	36.	(3)
37.	(3)	38.	(3)	39.	(1)				
40.	A(r); B(q); C(p); D(s)								
41.	A(s); B(r); Cq); D(p)								
42.	A(p); B(q); C(r); D(s)								
43.	A(s); B(q); C(r); D(q)								
44.	4	45.	3	46.	(1)	47.	(1)	48.	(1)
49.	(4)	50.	(1)	51.	(3)	52.	(1, 3, 4)	53.	(1, 2)
54.	(2)	55.	(4)	56.	(4)	57.	(4)	58.	(2)
59.	(2)								
60.	A(q); B(r); C(p); D(s)								
61.	A(p); B(t); C(q); D(s)								
62.	2	63.	(3)	64.	(3)	65.	(2)	66.	(1)
67.	(1)	68.	(2, 3, 4)	69.	(1, 2, 3)	70.	(3)	71.	(4)
72.	(2)	73.	8	74.	6	75.	1	76.	(1)
77.	(3)	78.	(3)	79.	(2)	80.	(2, 3, 4)	81.	(2, 4)
82.	(4)	83.	(1)	84.	(2)				
85.	A(p); B(p, q); C(p, q, r, s); D(s)								
86.	A(s); B(r); C(p, s); D(q, s)								
87.	40								